2nd Year Test

**Name:** Oisín Argand

94/100 excellent Oisín! you prepared really well for this.

505

**Q1.**

Write in the form $2^k, k \in Q$

i) 1

ii) 32

iii) $\sqrt{8}$

(i) 1 $2^0$ (ii) 32 $2^5$ (iii) $\sqrt{8}$ $2^{3\frac{1}{2}}$

5

**Q2.**

Simplify give your answer in the form $3^k,\ k \in Q$

i) $(3^2)^{-3}$

$(3^2)^{-3}$
$3^{-6}$

5

ii) $\frac{3^7}{9}$

$\frac{3^7}{9}$

$\frac{3^7}{3^2} = 3^5$

5

(iii) $\frac{3^5 \times 3^{2/3}}{9^{\frac{1}{2}} \times 3^2}$

$\frac{3^5 \times 3^{2/3}}{9^{\frac{1}{2}} \times 3^2}$ $\frac{3^{5\frac{2}{3}}}{9^{\frac{1}{2}} \times 3^2}$ $\frac{3^{5\frac{2}{3}}}{3^{2.05} \times 3^2}$ $\frac{3^{5\frac{2}{3}}}{3^{4.05}}$ $3^{1.616}$

6/10

$9^{\frac{1}{2}} = \sqrt{9} = 3$

**Q3.**

**(a)**

At the start of 2025, in Tanzania, the population is estimated to be 69 606 995. Write this figure in the form $a \times 10^n$, where $1 \leq a \leq 10$, $n \in \mathbb{N}$. Give your answer to 2 decimal places.

$6\,9\,6\,0\,6\,9\,9\,5$

$6{\cdot}96 \times 10^7$

✓ 5

**(b)**

The land area of Tanzania is approximately $9 \cdot 45087 \times 10^5$ square kilometres. Write this as a natural number.

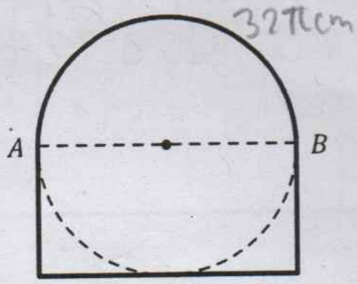
$9{\cdot}45087 \times 10^5$

945,087

✓ 5

**Q4.**

The image below shows a wooden window shutter. The shutter is made up from a semi-circle and a rectangle as shown on the diagram on the right. The length of the arc from $A$ to $B$ is $32\pi$cm.




**(a)** Find the **perimeter** of the full window, correct to two decimal places.

$32\pi cm$

$32\pi cm = 2\pi r \times \frac{180}{360}$

$32cm = 2r \times \frac{180}{360}$

$64cm = 2r$ $\quad 2\pi(32) \times \frac{180}{360}$

$32cm = r$ $\quad 100.53$

$2\pi r \times \frac{180}{360}$ $\quad 32 \times 2 + 64$

$128$

$100.53 + 128$

$\underline{228.53cm}$ ✓ excellent 10

**(b)** Find the **area** of the full window, correct to three decimal places.

$32\pi cm$

$32\pi cm = 2\pi r \times \frac{180}{360}$ $\quad \pi r^2 \times \frac{180}{360}$

$32 cm = 2r \times \frac{180}{360}$ $\quad \pi(32)^2 \times \frac{180}{360}$

$64 cm = 2r$ $\quad 1608.495$

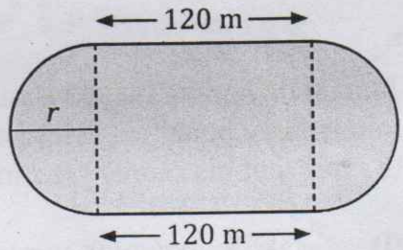
$32 cm = r$ $\quad 32 \times 64 = 2048$

$2048 + 1608.495$

$\underline{3656.495 cm^2}$ ✓ 10

**Q5.**

A greyhound track is designed as a rectangle with semi-circles on both ends. The **total** length of the track is 480 metres. The straight sections of the track are each 120 metres long.




**(a)** Calculate the radius of the semi-circles.

Give your answer to 1 decimal place.

$480 - 120 \times 2$

$240$

Circumfrence $= 240$

$240 = 2\pi r$

$120 = \pi r$

$\underline{38.2m = r}$ ✓ 10

**(b)** Find the area of the track. Give your answer to 2 decimal places.

38.2x2
76.4
76.4x120
9168

$\pi r^2$
$\pi(38.2)^2$
4584.34

9168+4584.34
13752.34 m²

10

**Q6.**

In this question, each shape has a fixed value.

**(a)** **(i)** Work out the value of a circle, using the following equation:

● + ● + ● = 21

● = 7

21 ÷ 3
7

**(ii)** Work out the value of a triangle, using the following equation, and your answer to part **(a)(i)**:

● + ▲ + ● = 18

5

18 − 7x2
4

▲ = 4

**(b)**

John and Gemma played a new computer game called *Benga*. John scored two bengas minus three penalties. His total score was seven points. He made the equation $2x - 3y = 7$ to represent his score. Gemma scored five bengas minus five penalties for twenty points.

**(i)** Make an equation to represent Gemma's score.

$5x - 5y = 20$

5

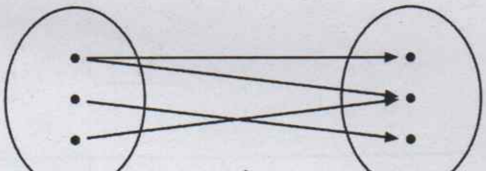
**Q7.**

**(a)** Solve the equation $13 - 3(x - 3) = 49$.

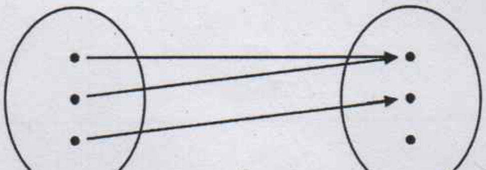
$13 - 3(x - 3) = 49$
$13 - 3x + 9 = 49$
$13 - 3x = 40$
$-3x = 27$
$3x = -27$
$x = -9$

10

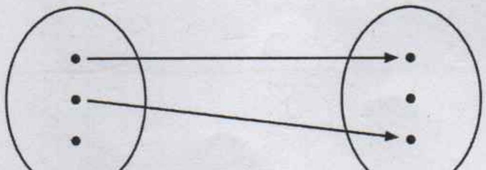
**(b)** One of the three arrow diagrams below is a **function.** Identifty which one is a function and say why the others are **not** a function.



*A*



*B*



*C*

5

A is not a function because an input cannot have multiple outputs ✓

B is a function because every input gets one output ✓

C is not a function because not every input gets an output ✓